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Course Outline

Name of the Program: Bachelor of Science in Computer Science and Engineering
Name of the Faculty: Electrical and Computer Engineering (ECE)
Name of the Course Offered Department: Computer Science and Engineering (CSE)
Name of the Course Conducting Department: Mathematics

Program:	Bachelor of Science in Computer Science and Engineering (B.Sc in CSE)
Course Code & Title:	MATH-2241 (Complex Variable and Fourier Analysis)
Level/Year:	2nd Year 2nd Semester
Credit Hour:	3.00 Hr
Name & Designation of Course Teacher:	Maksudul Hasan Likhon, Lecturer, Department of Mathematics
Class Hour:	Sunday:
Consultation Hour:	Sunday: Monday:
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Rationale:	The Course on Complex Variables and Fourier analysis offers essential knowledge and skills for students in engineering. Complex variables provide a critical foundation for understanding and solving advanced mathematical problems, including differential equations and integrals. These concepts are pivotal in various practical fields such as computer and electrical engineering, where they simplify in algorithm, graphics and cryptography often utilises complex variables as well as the analysis of alternating current circuits and signal processing. Fourier analysis is a powerful mathematical tool that enables the decomposition, analysis, and manipulation of signals by transforming them into the frequency domain. Its ability to reveal the frequency content of signal, combined with its application in signal processing, communication data analysis and engineering, makes it indispensable in modern science and technology. Fourier analysis provides an essential framework for solving practical problems in a wide variety of fields, from audio and image processing to the modelling of physical systems.

	The combination of complex variable analysis and Fourier analysis is powerful because both are based on similar mathematical principles and have overlapping applications. The ability to represent signals and functions in terms of complex essential (through the Fourier transforms) relies on complex analysis techniques. These methods together provide essential tools for solving problems in a range of disciplines, from engineering and physics to applied mathematics and computer science. The synergy between these areas of study allows for more efficient problem solving, deeper insights and better understanding of complex systems in both theoretical and applied contexts.
Pre-requisite (if any):	1. **Calculus (Differential & Intregal):** Essential for understanding integration and differentiation techniques. **Single-variable Calculus:** Limits, continuity, differentiation and integration techniques of integration as including integration by parts, substitution, and partial fractions, the fundamental theorem of calculus and applications of integration. **Multivariable Calculus:** Partial derivatives, multiple integrals and vector calculus, Green's theorem, Stokes' theorem and Divergence theorem, Gradient, divergence, curl and their physical interpretations (important for complex analysis in physics applications). 2. **Differential Equations (Ordinary & Partial):** Critical for solving and applying solutions to differential equations using the methods taught in the course. **Key Topics:** Solving first-order and high-order ODEs, methods for solving linear differential equations (both homogenous and non-homogeneous), Laplace transforms and their relationship to Fourier transforms, solutions of PDEs, particularly those related to wave, heat and Laplace equations (important for physical applications of Fourier analysis).
Course Objectives:	**<u>Objectives of Complex Variable Analysis</u>** 1. **Understanding Complex Functions and Their Properties:** Develop a understanding of functions of a complex variable, focusing on their properties like analyticity (holomorphic functions), singularities and branch points. 2. **Mastering Complex Differentiation and Integration:** Learn the methods of complex differentiation and integration, including understanding Cauchy-Riemann equations and using contour integration to evaluate integrals. 3. **Utilising the Cauchy-Riemann Equations:** Learn how to apply the Cauchy-Riemann equations to verify whether a function is differentiable in the complex plane and study properties of such functions. 4. **Applying the Residue Theorem:** Use the Residue Theorem for evaluating complex integrals, especially those that are difficult or impossible to compute using real analysis methods. 5. **Exploring Conformal Mappings:** Study and apply conformal mappings to transform complex regions in way that preserves angles, which is especially useful in fluid dynamics and electrical engineering. 6. **Solving Physical Problems Using Complex Variables:** Apply techniques from complex analysis to solve problems in physics and engineering, including electromagnetism, fluid dynamics, and wave propagation.

7. **Classifying and Handling Singularities:** Learn to classify singularities (poles, essential singularities, branch points) of complex functions and understand their implications for function behaviour.
8. **Extending Functions Using Analytic Continuation:** Use analytic continuation to extend the domain of complex functions, allowing them to be evaluated in regions beyond their original domain.

Objectives of Fourier Analysis

1. **Understanding_Fourier Series:** Learn to express periodic functions as sums of sine and cosine (Fourier Series), and understand the conditions for the convergence of the series.
2. **Transforming Signal into Frequency Domain:** Develop the ability to transform time domain signals into the frequency domain using Fourier transforms, gaining insight into their frequency content.
3. **Analysing Non-Periodic Functions with Fourier Transforms:** Apply Fourier transforms to non-periodic signals and understand how they reveal the continuous spectrum of a signal.
4. **Mastering the Discrete Fourier Transform (DFT):** Understand and implement the Discrete Fourier Transform (DFT) and use the Fast Fourier Transform (FFT) algorithm for efficient computation in digital signal processing.
5. **Applying Fourier Analysis in Signal Processing:** Use Fourier Analysis in practical applications like filtering, noise reduction and compression, particularly in fields such as audio, image, and video processing.
6. **Solving Partial Differential Equations Using Fourier Methods:** Use Fourier Series and transforms to solve partial differential equations (PDEs) commonly found in physics such as heat conduction, wave propagation and vibration analysis.
7. **Studying System Behaviour in the Frequency Domain:** Use Fourier analysis to study frequency response of systems, allowing for design of filters, amplifiers and other components in engineering and communications.
8. **Understanding Signal Representation and Data Compression:** Use Fourier analysis to efficiently represent signals, compress data and analyze the frequency components of signals, particularly in telecommunications and media compression.

Combined Objectives of Complex Variable and Fourier Analysis

1. **Develop a Deep Understanding of Complex Functions and Their Applications:** Gain Proficiency in Analysing and solving problems involving complex functions, including holomorphic (analytic) functions, singularities and residues.
   Apply these concepts to enhance the understanding of Fourier analysis, particularly when dealing with complex exponentials and the frequency domain.

2. **Master the Use of Fourier Transforms and Series in Complex Plane:** Understand how Fourier transforms can be formulated and analysed using complex variables, particularly with the use of complex exponentials (Euler's Formula).
   Apply the properties of complex functions to the Fourier transform and Fourier series to solve real world problems in both continuous and discrete Settings.

3. **Solve Real-World Problems in Signal Processing and Communication Systems:** Use the tools from both complex analysis and Fourier analysis to design and optimise signal processing algorithm, including filtering, compression and noise reduction.
   Analyse communication systems in the Frequency domain using complex Variables, Fourier transforms and series to better understand modulation Transmission and reception of signals.

4. **Understand and Apply the Relationship Between Time and Frequency Domains:** Leverage the duality between time-domain and frequency-domain representations of signals. Use Fourier analysis to decompose signals into their frequency components and apply complex analysis techniques to study their properties.
   Gain insights into behaviour of signals and systems in both time and freq-ency domains, enabling more efficient problem solving engineering and Physics.

5. **Solve the Differential Equations using Fourier And Complex Analysis Methods:** Apply Fourier methods (series and transforms) alongside complex variable techniques to solve partial differential equations (PDEs) and ordinary differential equations (ODEs) that arise in physics and engineerings such as heat, wave and Laplace equations.
   Use complex integration and the residue theorem to evaluate integrals th-at arise when solving differential equations.

6. **Analyse and Design Systems Using Frequency-Domain Techniques:** Use the frequency-domain analysis tools from Fourier analysis to study the behaviour and stability systems in engineering, control theory and physics.
   Apply complex analysis to understand system responses such as resona-nce, damping filtering in circuits, mechanical systems and signal proces-ing.

7. **Optimize Computational Methods for Signal and Systems Analysis:** Develop efficient computational methods by combining Fourier analysis (e.g. Fast Fourier Transform or FFT) with complex analysis techniques to handle large datasets in digital signal processing. Use complex-variable-based techniques for efficient numerical integration and and the evaluation of complex integrals, particularly engineering applications.

8. **Apply Advanced Mathematical Techniques to Physical Systems:** Use the combined framework of complex variable and Fourier analysis to model and analyse physical phenomena, such as wave propagation, electromagnetic fields, quantum mechanics and fluid dynamics.
   Solve physical problems involving boundary value problems, Green's functions and wave equations using Fourier and complex analysis methods.
9. **Provide a Unified Mathematical Approach to Complex Problems:** Integrate the principles of complex variable theory and Fourier analysis to provide a unified approach to solving both theoretical and applied problems in mathematics and engineering.
   Explore the interplay between the two fields to create more sophisticated solutions to problems in fields like optics, acoustics and electrical engineering.
10. **Lay the groundwork for Advanced Research in Mathematical and Applied Sciences:** Build a foundation for further study in advanced fields such as functional analysis, quantum mechanics and computational mathematics by mastering complex variable and Fourier analysis techniques.

	Prepare for research and development in emerging technologies such as signal processing, image processing, machine learning and computational physics, where these mathematical tools are indispensable. **Together, these objectives help students and practitioners apply the combined power of complex variable analysis and Fourier analysis to solve real-world challenges in various scientific and engineering disciplines**
Course Outcomes (CO):	Course Outcomes of this course are: **CO1: Understanding and Analysing** Students will be able to understand and explain key concepts related to complex variable and Fourier analysis. **CO2: Problem-Solving Skills** Students will apply complex variable theory and Fourier analysis to enhance their problem-solving abilities as well as evaluate and interpret the solutions in the context. **CO3: Applications** Students will demonstrate their understanding and applications of complex variables and Fourier analysis in engineering problems. These outcomes ensure that students not only learn the the theoretical aspects of complex variables and Fourier analysis but also develop the skills to apply these concepts in practical and real-world contexts.
Course Content:	**Complex Variable** Complex number system, complex functions, limits & continuity of complex variables and related theorems, complex differentiation & Cauchy-Riemann equations, analytic & harmonic functions, singularities, line integral of complex functions, Cauchy's integral theorem, Cauchy's integral formula, Liouville's theorem, Taylor's theorem and Laurent's theorem, residues and Cauchy's residue theorem, contour integration and conformal mapping. **Fourier Analysis** Real and complex form of Fourier series, Fourier integral, Fourier transforms, inverse Fourier transforms and their uses in solving boundary value problems. **Text Books:** 1. Murry R. Speigal, Seymour Lipschitz, John J. Schiller and Dennis Spellman, Complex Variables, Schaum's Outline Series, McGraw Hill, 2010, 2nd Edition, ISBN: 978-0-07-008538-1 2. Murray R. Spiegel, Fourier Analysis with applications to Boundary Value Problems, Schaum's Outline Series, McGraw Hill, 1965, 2nd Edition. **Reference Books:** **1.** James Ward Brown and Ruel V. Churchill, Complex Variables and Applications, SEM, 2013, 9th Edition, ISBN-13: 978-0073383170 **2.** M.D Raisinghania, Laplace and Fourier Transformations, S. Chad & Company Ltd, 1995, ISBN: 8121908841 9788121908849 **3.** Complex Analysis- A.K.M. Shahidullah **4.** Complex Analysis-Prof Dr. M.F. Rahman **5.** The Elements of Complex Analysis-B. Choudhary **6.** Methods of Applied Mathematics- Prof. Md. Abu Yousuf **7.** H. K. Das, Advanced Engineering Mathematics, S. Chand & Company Ltd **8.** Erwin Kreyszin, Advanced Engineering

Student's Responsibilities:	• Students must come to the class prepared for the course material covered in the previous class(es). • They must submit their assignments on time. • No late or partial assignments will be acceptable. • There will be no make-uo class test.

Teaching and Learning Strategies (TLS)

TLS1	Delivery lectures using whiteboard/multimedia of different topics
TLS2	Active and fruitful discussion in class regarding efficient solving of logical, mathematical and complex engineering problems.
TLS3	Group discussion and presentation regarding diverse problems and corresponding lectures
TLS4	Evaluation of class performance to reach each student in a class for every topic.

List of Content with Lecture Plan

Lecture SL	Topic SL	Contents	Week
		Complex Variable	
	1	Complex Number System	
1		a. Definitions	W1
2		b. Properties	W1
3		c. Problems (General)	W1
4		d. Problems (Roots)	W2
5		e. Problems (Sketch the region in z-plane represented by the given relation)	W2
6	2	General Functions of a complex variable	W2
	3	Limits and continuity of functions of complex variable and related theorems	
7		a. Limits of functions of complex variables and related theorems	W3
8		b. Continuity of functions of complex variables and related theorems	W3
9		**Class Test-1 and Discussion**	W3
	4	Complex differentiation and the Cauchy-Riemann equations	
10		a. Complex differentiation	W4
11		b. Cauchy-Riemann equations (Theorem and Properties)	W4
12		c. Cauchy-Riemann equations (Problems)	W4

13		d. Analytic functions	W5
14		e. Harmonic function	W5
15	5	Singularities	W5
16		**Class Test-2 and Discussion**	W6
17	6	Line Integral of Complex function	W6
18	7	Cauchy’s Integral theorem	W6
19	8	Cauchy's integral formula	W7
	9	Liouville’s theorem	
20		a. Liouville’s theorem b. Liouville’s problems	W7
21		**Assignment**	W7
		Midterm Examination	
	10	Taylor’s Theorem and Laurent’s theorem	W8
22		Taylor's theorem and problems	W8
23		Laurent’s theorem and problems	W8
24	11	Singular points	
	12	Residues and Cauchy’s Residue theorem	
25		a. Residues	W9
26		b. Cauchy’s Residue theorem	W9
27		c. Evaluation of residues	W9
28	14	Contour Integration	W10
29		Contour Integration	W10
30	15	Conformal mapping and Review Class (Discussion)	W10
Total Class(Complex Variable): 30			
		Fourier Analysis	
	1	Fourier Series	
31		a. Real form and properties	
32		b. Real form and problems	
33		c. Complex form and properties	
34		d. Complex form and problems	
35		**Class Test-3 and Discussion**	

	3	Fourier Integral	
36		a. Fourier Integral and properties	
37		b. Fourier Integral and problems	
	4	Fourier transforms	
38		a. Fourier transforms and properties	
39		b. Fourier transforms and problems	
40	5	Inverse fourier transforms	
41	6	Solving Boundary Value Problems (BVP) by fourier transforms	
42		**Class Test-4 and Review Class (Discussion)**	
Total Class (Fourier Analysis and Review Class): 12			

Summary

Total Class: 37 (Complex Variable =27 and Fourier Analysis=10)
Total Class Test and Review Class (Discussion): 4 (Complex Variable=2 and Fourier Analysis=2; Best 3 out of 4 Class Tests/Assignments will be considered)
Total Assignment: 1
Total Conduct Hours: 42 Hours
Grand Total Class: 42 (Complex variable-30; Fourier Analysis-12)